

Lab Exercise 18

HIGH LEVEL LANGUAGE EXTENSION WITH CURSORS

AIM

To implement an **explicit cursor** in MySQL to retrieve vegetable names from the `Vegetable` table and build a list.

PROCEDURE

1. Select the `VegetableDB` database.
2. Declare a cursor to fetch `VegName` from the `Vegetable` table.
3. Declare a handler to detect the end of the cursor.
4. Open the cursor.
5. Fetch each vegetable name row by row.
6. Add each name to a list.
7. Close the cursor.
8. Display the final vegetable list.

SQL PROGRAM

1. Select Database

```
USE VegetableDB;
```

2. Create Cursor Procedure

```
DELIMITER //
```

```
CREATE PROCEDURE BuildVegetableList(INOUT veg_list VARCHAR(4000))
BEGIN
    DECLARE finished INT DEFAULT 0;
    DECLARE v_name VARCHAR(30);

    DECLARE veg_cursor CURSOR FOR
        SELECT VegName FROM Vegetable;

    DECLARE CONTINUE HANDLER FOR NOT FOUND
        SET finished = 1;

    OPEN veg_cursor;

    get_vegetable: LOOP
```

```

        FETCH veg_cursor INTO v_name;

        IF finished = 1 THEN
            LEAVE get_vegetable;
        END IF;

        SET veg_list = CONCAT(veg_list, v_name, ';');

    END LOOP get_vegetable;

    CLOSE veg_cursor;
END //

DELIMITER ;

```

3. Call the Procedure

```

SET @veg_list = '';

CALL BuildVegetableList(@veg_list);

SELECT @veg_list AS Vegetable_List;

```

```

mysql> SELECT @veg_list AS Vegetable_List;
+-----+
| Vegetable_List |
+-----+
| Tomato;Potato;Carrot;Onion;Beetroot;Cabbage;Spinach;Beetroot; |
+-----+
1 row in set (0.00 sec)

```